Job Projections and AI Automation

Summer 2026

C1 - Portfolio Project DATA CURATION

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General Sources/ Dataset Information

#1

File Name: occupation.xlsx

Description:U.S Bureau of Labor Statistics 2024

Dataset Details: 1119 Rows & 16 Columns

Size: 398 KB

Source: (U.S) Bureau of Labor Statistics - Occupational projections, 2024–2034, and worker characteristics, 2024 (Employment in thousands)

#2

AIOE Dataset *

FIle Name: AIOE_DataAppendix.xlsx

Description: AIOE Data- Appendix A

Dataset details: 775 rows & 3 columns

Size: 167 KB

Source: GitHub- AIOE Data

* AIOE stands for Artificial Intelligence Occupational Exposure.

Data Profile/ Process

- Downloaded occupation.xslx dataset from USBL site to chromebook. Uploaded dataset to Excel online.
- The original dataset consisted of 13 sheets. Each sheet is a breakout subset of table 1.2
- Table 1.2 is to be used as master source/tab and accordingly was copied and pasted into a blank workbook for purposes of cleaning.

- Line 1 and line 3 in table 1.2 were summaries of table 1.2.'s contents- will be deleted to reduce redundancy.
- Column C cells reading " summary" in table 1.2 were filtered out/ deleted to avoid redundancy. "Summary" lines repeat what is to be analyzed as line items further.
- Columns D-L in 1.2 are to be changed from general to numeric categories for cleaning & wrangling efficiency.
- Rows 1116-1119 will be deleted because they contain footnotes. Columns N-P were deleted in table 1.2 because they are non-numeric, not helpful in the context of analysis.
- Located AIOE dataset; Appendix A is to be merged with Bureau of Labor Statistics data.
- AIOE data came readily available. Appeared correctly formatted. Pasted dataset into a new sheet in the same workbook as Bureau Labor Statistics table 1.2 for merging.
- Used Index-Match function to Merge AIEO scores into new column O of table 1.2.
- Deleted heart row of occupation table to prompt straightforwardness.

Data Cleaning

- Index -match processed - appendix A AIOE scores into table 1.2 as new column D.
- Some detailed occupations don't have an original AIOE score, they have blanks.
- Instead of leaving those cells blank, they were estimated using the average score of similar occupations. First attempt: 4-digit SOC family
- If that family has no score, it will use the broader 2-digit SOC group.

Final data table contains : 14 columns, 832 rows in final data set

Data Table/ Schema

Column name	Description
Employment title	The name of the occupation represented in the dataset. This field identifies the specific career or occupational group being analyzed.
2024 National Employment Matrix (SOC) code	The Standard Occupational Classification (SOC) code assigned to each occupation by the BLS. It provides a standardized identifier that can also be used to match occupations across datasets.
Occupation Type	Indicates how the occupation is categorized with BLS Projections table, such as a detailed line-item occupation or occupational group.
AIOE	Measures how strongly an occupation's tasks are exposed to advances in AI . Higher scores indicate greater AI exposure, lower scores indicate less AI exposure.
Employment, 2024	Estimated number of people in the occupation in 2024. BLS reports employment figures in thousands of jobs.
Employment, 2034	Projected number of people in the occupation in 2034.
Employment distribution, percent, 2024	% of total U.S employment represented by occupation in 2024. It provides context for the occupation's relative size in the labor market.
Employment distribution, percent, 2034	Projected % of total U.S employment represented by occupation in 2034. It shows whether the occupation's relative share of employment is expected to change.
Employment change, numeric, 2024–34	Projected numeric change in employment from 2024-2034, reported in thousands of jobs. "+" values indicate job gains.

Employment change, percent, 2024–34	Projected % increase or decrease in employment from 2024-2034. This variable is especially useful for comparing expected job growth across occupations of different sizes.
Percent self employed, 2024	% of workers within an occupation who were self-employed in 2024. This provides insight into how much an occupation relies on independent or self-directed employment.
Occupational openings, 2024–34 annual average	Avg number of job openings expected each year from 2024-2034. Includes openings created by employment growth & need to replace workers who leave occupation.
Median annual wage, dollars, 2024[1]	Median annual wage for workers in the occupation in 2024. This variable helps evaluate the economic value and earning potential of each occupation.
Typical education needed for entry	Typical level of education workers needed to enter the occupation such as a bachelor's, or if no formal educational credential. This helps assess the educational requirements associated with different career occupations.